

EXACT ACTION SEPARATION ON THE MAIN COMPONENT OF A PRETZEL-TANGLE COMPOSITION IN THE PILLOWCASE

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ABSTRACT. Let $P = (S^1 \times S^1)/\{(\gamma, \theta) \sim (-\gamma, -\theta)\}$ be the pillowcase and let P^* be its smooth stratum. We compose the rational-tangle arcs for $Q_{1/3}$ and $Q_{1/7}$ with the small-holonomy-perturbation correspondence of the three-strand tangle. Smith's cross-resolution distinguishes a main output, obtained by resolving six internal circle fibers and continuing the branches away from them. After a symmetric compact truncation, we prove that the correspondence is exact and that a half-period symmetry fixes the relative primitive on the two components of this main output.

In the singular limit the main output consists of 21 affine branches joined across the six circle fibers. Their twelve endpoints have nondegenerate hyperbolic smoothing models. The branch-intersection equations have 58 limiting candidates, exactly 50 of which continue to transverse ordered self-intersections for every sufficiently small positive perturbation. An area-residue calculation evaluates each limiting action as a rational multiple of π^2 : 42 are negative and 8 are positive, no action vanishes, and the smallest absolute limiting action is $\pi^2/42$. The count and all action signs persist under perturbation. No claim is made about auxiliary circle components allowed by the cross-resolution, the noncompact puncture ends, holomorphic-quilt counts, bounding cochains, or the instanton-pillowcase comparison.

1. INTRODUCTION

The traceless $SU(2)$ character variety of the four-punctured sphere is the pillowcase

$$P = (S^1 \times S^1)/(\gamma, \theta) \sim (-\gamma, -\theta).$$

Its smooth stratum P^* is obtained by deleting the four orbifold points; we use the standard Liouville completion near those punctures [CHKK22, Appendix A.1.1]. Rational tangles give proper Lagrangian arcs $A_r \subset P^*$, while a holonomy perturbation of the three-strand tangle gives an immersed Lagrangian correspondence

$$C_t = C_{3,t}^{\text{reg}} \looparrowright (P_1^*)^- \times (P_2^*)^- \times P_3^*$$

for small $t > 0$ [HK18, Smi24]. We first form the full direct fiber product

$$\tilde{L}_t^{\text{full}} = (A_{1/3} \times A_{1/7}) \times_{P_1 \times P_2} C_t, \quad L_t^{\text{full}} = \pi_3(\tilde{L}_t^{\text{full}}) \looparrowright P_3^*. \quad (1)$$

Smith's cross-resolution theorem singles out a main subspace obtained by resolving the internal circle fibers and continuing the off-circle branches; it also permits auxiliary circle components that collapse toward isolated singular images as $t \rightarrow 0$ [Smi24, Thm. 4.22]. Fix a generic symmetric truncation $P^\delta \subset P^*$ containing the six circle images and all intersections studied below. We write

$$\tilde{L}_t^\delta \subset \tilde{L}_t^{\text{full}}, \quad L_t^\delta = \pi_3(\tilde{L}_t^\delta) \looparrowright P^\delta$$

for the compact main subspace and its output. Auxiliary components and the portions in the deleted end collars are not included in L_t^δ .

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The pair of slopes $1/3$ and $1/7$ is the character-variety composition associated to the pretzel-tangle presentation of $P(-2, 3, 7)$ used in the companion calculation [Wue26].

The purpose of this paper is finite and geometric. It determines the exact immersed-curve action data of the compact main output L_t^δ ; it does not classify the auxiliary components of (1) or construct any moduli space of figure-eight bubbles. This distinction matters because strip shrinking can produce Morsified trees of several holomorphic components [BW18]. A list of candidate output generators is not a count of those trees.

Following Smith, we call the loci $\gamma = 0$ and $\gamma = \pi$ in an input pillowcase the *seams*; these are not seams of a pseudoholomorphic quilt. In the singular limit, the output branches are either affine graph segments or images of the six circle fibers. We call an intersection of the two types a *diagonal-circle intersection*, and an intersection of two circle images a *circle-circle intersection*.

We fix the remaining terminology. If a transverse self-intersection of L_t^δ has two preimages z_- and z_+ , the orientation convention for immersed Floer theory selects one ordered pair $x = (z_-, z_+)$ in degree one. We call this ordered generator a *root*. If f_t is the primitive inherited from the exact fiber product, its action is

$$\mathcal{A}_t(x) = f_t(z_-) - f_t(z_+). \quad (2)$$

The word “root” in this paper therefore means an ordered branch jump, not a zero of an unspecified function. The *short deck class* of a root is the reduced torus-deck displacement of the ordinary closing loop used in the area-residue calculation; for an odd seam separation, the construction first doubles the path and then removes that artificial factor of two.

Theorem 1.1 (Exact action theorem). *There are an equivariant Liouville completion of the pillowcase, a generic symmetric truncation P^δ , and a number $t_0 > 0$ with the following properties for every $0 < t < t_0$.*

- (1) *The compactly truncated correspondence C_t^δ is exact. The main output L_t^δ inherits a primitive whose relative normalization across its two open parameter arcs is fixed by a half-period symmetry.*
- (2) *The compact main output L_t^δ has exactly 50 transverse degree-one roots arising from the singular composition: 28 diagonal-circle roots and 22 circle-circle roots. They occur in 25 half-period pairs.*
- (3) *The limiting actions as $t \rightarrow 0^+$ are nonzero rational multiples of π^2 . Exactly 42 roots have negative limiting action and 8 have positive limiting action. The minimum absolute limiting action is $\pi^2/42$, and all 50 action signs persist throughout $0 < t < t_0$.*

The four positive half-period representatives have short deck classes and limiting actions

<i>seam pair</i>	<i>short deck</i>	$\lim_{t \rightarrow 0^+} \mathcal{A}_t / \pi^2$
(1, 7)	(3, 1)	3/7
(2, 7)	(8, 4)	5/14
(2, 17)	(3, 1)	3/14
(3, 4)	(10, 5)	5/42.

The theorem concerns the main subspace L_t^δ . It neither asserts that Smith’s possible auxiliary circle components are absent from L_t^{full} nor assigns actions to them.

Three ingredients make the statement exact. First, the Liouville primitive is normalized equivariantly and its periods are checked on the full singular fiber, including seam-born and corner-link cycles. Hamiltonian-composition flux then transports this vanishing to $t > 0$. Second, the singular output is recorded by exact piecewise-linear first-arc data. Its only failures of the graph description occur at six explicitly determined stabilizer circles, whose endpoints carry hyperbolic smoothing germs. Third, all circle-circle intersections reduce to quadratic equations

over $\mathbb{Q}(2\cos(\pi/7))$. Rational interval bounds isolate the physical solutions and determine their derivative signs. The action of each surviving root is then a signed area minus a lattice-residue term.

Section 2 fixes the primitive and proves exactness. Section 3 describes the six folds, the hyperbolic smoothing, and the 21 separated graph sheets. Section 4 proves the exact root count and action table. Section 5 records the precise energy consequence and the boundary of the result.

2. EXACT PRIMITIVES AND HALF-PERIOD NORMALIZATION

Write (γ, θ) for the torus-cover coordinates on the pillowcase. On the central cylinder

$$U = (0, \pi) \times S^1 \subset P^*,$$

the Goldman form is $d\gamma \wedge d\theta$. The choice of a global primitive on the completed four-punctured sphere includes residue data at its four ends.

Lemma 2.1 (Normalized pillowcase completion). *Let*

$$\tau[\gamma, \theta] = [\gamma, \theta + \pi], \quad \sigma[\gamma, \theta] = [\pi - \gamma, -\theta].$$

There is a $\langle \tau, \sigma \rangle$ -invariant primitive λ_0 of the Goldman form on P^ such that*

$$d\lambda_0 = d\gamma \wedge d\theta, \quad \lambda_0 = (\gamma - \pi/2) d\theta$$

near the circle $\{\gamma = \pi/2\}$. Its four positively oriented puncture residues are all $-\pi^2/2$. For every sufficiently small invariant corner truncation $P^\delta \subset P^$, there is an invariant Liouville completion $(\hat{P}^\delta, \lambda)$ whose primitive agrees with λ_0 on P^δ .*

Proof. The noncompact surface P^* has zero second cohomology, so its Goldman form has a primitive λ' . On an annular neighborhood of $\{\gamma = \pi/2\}$, add a closed one-form to kill the period of $\lambda' - (\gamma - \pi/2)d\theta$, and then subtract the differential of a cutoff of the resulting local function. Average the primitive and the closed correction over $\langle \tau, \sigma \rangle$. Both the Goldman form and the displayed central primitive are invariant. The group acts transitively on the four punctures, so the four residues agree.

The torus cover has area $4\pi^2$ and is generically two-to-one over P , hence the Goldman area of P^* is $2\pi^2$. Apply Stokes' theorem to an invariant exhaustion by compact surfaces. With a counterclockwise meridian around a deleted point declared positive, the sum of the four limiting meridian periods is the negative of this area. Equality of the residues therefore makes each one $-\pi^2/2$. The averaged closed correction does not alter them: its four residue changes are equal and have sum zero.

Finally choose the four deleted disks equivariantly. Outside the prescribed compact truncation, modify the primitive in disjoint collars so that its Liouville vector field is outward transverse, and attach the positive symplectizations of the boundary circles. These changes can be made equivariantly and leave λ_0 unchanged on P^δ . $\square$

Put

$$\Lambda_P = -\lambda_1 - \lambda_2 + \lambda_3.$$

On a smooth torus-cover chart of the unperturbed correspondence, Smith's boundary coordinates are

$$\gamma_1 = \gamma_2 = \gamma_3 = \gamma, \quad \theta_1 = \beta \sin \alpha, \quad \theta_2 = \theta - \beta \sin \alpha, \quad \theta_3 = \theta \quad (3)$$

[Smi24, Eqs. (3.1.4)–(3.1.5)]. At $\gamma = \pi/2$, the normalized local primitive therefore satisfies $\Lambda_P|_{C_0} = 0$.

Let

$$\mathcal{V}_0 = \{\sin \gamma \cos \alpha = 0\}, \quad r_0 = (r_1, r_2, r_3) : \mathcal{V}_0^\delta \longrightarrow (P^\delta)^3$$

be Smith's singular parameter fiber and its three boundary maps. If c is an oriented loop in $\mathcal{V}_0^\delta$, write

$$h(c) = -r_{1*}[c] - r_{2*}[c] + r_{3*}[c] \in H_1(P^*; \mathbb{Z}).$$

Proposition 2.2 (Exactness of the small perturbation). *For a generic symmetric truncation P^δ and every sufficiently small $t > 0$,*

$$[\ell_t^* \Lambda_P] = 0 \quad \text{in } H^1(C_t^\delta; \mathbb{R}),$$

where $\ell_t : C_t^\delta \hookrightarrow (P^\delta)^- \times (P^\delta)^- \times P^\delta$ is the restriction immersion. The assertion includes the cycles created when Smith's singular seam is smoothed and all corner-link cycles.

Proof. We first calculate on the regular part of the singular parameter fiber. Smith's parameter space $\mathcal{V}_0^*$ is a two-fold cover of the regular correspondence and decomposes as

$$\mathcal{V}_0^* \cong [S^2 \times (S^1 \setminus S^0)] \sqcup [T^2 \times (S^1 \setminus S^0)]$$

[Smi24, Def. 3.6]. Each component of the first summand has zero first real cohomology. On a component of the second summand, represent the two torus generators at $\gamma = \pi/2$. Equation (3) gives angular increments $(2\pi, -2\pi, 0)$ for the β -circle and $(0, 2\pi, 2\pi)$ for the θ -circle. The normalized primitive vanishes on both generators. The parameter involution exchanges the two components of $S^1 \setminus S^0$ and preserves the pulled-back one-form, so the period class vanishes on all of $\mathcal{V}_0^*$.

It remains to include cycles through the singular seam. Decompose the full zero set as

$$\mathcal{V}_0 = A \cup H_0 \cup H_\pi, \quad A = \{\alpha = \pi/2\} \cong T^3, \quad H_0 \sqcup H_\pi = \{\gamma = 0, \pi\} \cong S^0 \times S^1 \times S^2.$$

and put

$$A_\epsilon = \{\beta = \epsilon\pi\}, \quad B_\epsilon = \{\theta - \beta = \epsilon\pi\}, \quad \epsilon \in \{0, 1\}.$$

Cutting A along these four subtori and applying Mayer–Vietoris to the two $S^1 \times S^2$ pieces and the corner collars gives a redundant spanning set for first homology: the two circle directions on every A_ϵ and B_ϵ , together with the two directions on every corner link. The angular directions can be represented at $\gamma = \pi/2$ and have zero period by the preceding paragraph. Denote the remaining γ -directions by $A_0^\gamma, A_\pi^\gamma, B_0^\gamma, B_\pi^\gamma$, using boundary-collar push-offs when needed.

Orient the meridians $m_{00}, m_{0\pi}, m_{\pi 0}, m_{\pi\pi}$ positively about the four deleted points. Their sum is zero, so use $(m_{0\pi}, m_{\pi 0}, m_{\pi\pi})$ as a basis of $H_1(P^*; \mathbb{Z})$. Put $\mathcal{A}_\delta = \int_{P^\delta} d\lambda$. Direct substitution in the boundary formulas, followed by Stokes' theorem on the intervening rectangles in a pillowcase fundamental domain, gives

c	$h(c)$ in the displayed basis	$\int_c r_0^* \Lambda_P^{(00)}$
A_0^γ	$(0, 0, 0)$	0
A_π^γ	$(2, 0, 2)$	$\mathcal{A}_\delta$
B_0^γ	$(0, 1, -1)$	0
B_π^γ	$(-2, -1, -1)$	$-\mathcal{A}_\delta$.

(4)

Here $\lambda^{(00)}$ is a primitive whose positively oriented meridian periods are $(-\mathcal{A}_\delta, 0, 0, 0)$, and $\Lambda_P^{(00)} = -\lambda_1^{(00)} - \lambda_2^{(00)} + \lambda_3^{(00)}$. The middle column records the meridian crossed by each boundary projection as γ passes 0 or π ; it therefore prints every sign and multiplicity used in the comparison.

On the four boundary meridians of the symmetric truncation, the restricted primitive λ has period $-\mathcal{A}_\delta/4$ by Stokes' theorem and symmetry. Thus $\lambda - \lambda^{(00)}$ is closed and has period $-\mathcal{A}_\delta/4$ on each basis meridian. Pairing this period vector with the middle column changes the four entries in the final column by 0, $-\mathcal{A}_\delta, 0, \mathcal{A}_\delta$, respectively. All four γ -periods therefore vanish.

Index a parameter corner by $(g, h, k) \in \{0, 1\}^3$, meaning $(\gamma, \theta, \beta) = (g\pi, h\pi, k\pi)$. The value of k does not affect the following winding calculation. If (u, v) is an oriented basis of its torus link, the meridian winding triples in (P_1, P_2, P_3) are

(g, h)	$(w_1, w_2, w_3)(u)$	$(w_1, w_2, w_3)(v)$
$(0, 0)$	$(-1, 1, 0)$	$(1, 1, 2)$
$(0, 1)$	$(-1, -1, -2)$	$(-1, 1, 0)$
$(1, 0)$	$(1, 1, 2)$	$(-1, 1, 0)$
$(1, 1)$	$(1, -1, 0)$	$(1, 1, 2)$

(5)

Every vector satisfies $w_3 = w_1 + w_2$. Since the four end residues agree, the signed period of a link circle is the common residue multiplied by $-w_1 - w_2 + w_3$, and hence vanishes. The Mayer–Vietoris spanning set now shows that $r_0^* \Lambda_P$ has zero period on every class in $H_1(\mathcal{V}_0^\delta; \mathbb{R})$, including the classes lost when one looks only at $\mathcal{V}_0^*$.

For $t > 0$, cut out the perturbation solid torus. The perturbed correspondence is a geometric composition of a fixed restriction immersion with the solid-torus arc

$$[e^{is}] \mapsto [e^{is}, e^{ift(s)}],$$

which is a Hamiltonian deformation [CHK22, Eq. (6-4) and Sec. 6B]. If $v_t = \partial_t \ell_t$ is the vector field along the composed family, the Lagrangian condition gives

$$\ell_t^* \iota_{v_t} (-\omega_1 - \omega_2 + \omega_3) = -d(H_t \circ \text{pr}), \quad (6)$$

with the convention $\iota_{X_{H_t}} \omega = -dH_t$. Hence $\partial_t(\ell_t^* \Lambda_P) = dG_t$, where

$$G_t = \Lambda_P(v_t) - H_t \circ \text{pr}.$$

Equivalently, $\ell^* \Lambda_P + G_t dt$ is closed on the smooth trace of the family.

Let $q_t : \mathcal{V}_t^\delta \rightarrow C_t^\delta$ be Smith’s two-fold parameter cover and let $\mathfrak{V}^\delta$ be the closure of its compact trace over $0 \leq t \leq \epsilon$. Smith’s explicit equation and analytic boundary maps make $(\mathfrak{V}^\delta, \mathcal{V}_0^\delta)$ a compact subanalytic pair, hence a compatibly triangulable pair. A regular neighborhood N of the zero-fiber subcomplex retracts onto $\mathcal{V}_0^\delta$. After decreasing ϵ , the entire trace lies in N : otherwise a sequence outside N with $t \rightarrow 0$ would limit to a point of $\mathcal{V}_0^\delta \subset \text{int } N$. Consequently every one-cycle $\tilde{a}_t$ in a positive fiber is joined inside the total trace by a singular two-chain to a cycle a_0 in the full zero set; a vanishing cycle is the case $a_0 = 0$.

The Hamiltonian graph and all boundary maps extend analytically to $t = 0$, so the pulled-back trace form extends continuously and is smooth on every simplex stratum. Stokes’ theorem applied stratum by stratum gives

$$\int_{\tilde{a}_t} q_t^* \ell_t^* \Lambda_P = \int_{a_0} r_0^* \Lambda_P = 0.$$

Thus $q_t^*[\ell_t^* \Lambda_P] = 0$. Pullback on first real cohomology under a finite cover is injective by averaging over the deck group, which proves the proposition. $\square$

The rational arcs are contractible and can be normalized equivariantly, so $A_{1/3} \times A_{1/7}$ is exact. Primitives add under transverse geometric composition. Proposition 2.2 therefore gives a primitive on every component of the full direct fiber product (1), and hence on its compact main subspace. The main output, however, passes through two deleted corners in each lifted period. Its two open arcs initially have independent additive constants; the next proposition fixes their relative normalization.

Proposition 2.3 (Half-period normalization). *On the parameter cover of the direct composition, all-corner points on the main lift are*

$$z_j = (\gamma, \theta, \alpha, \beta) = (21j\pi, 10j\pi, \pi/2, 7j\pi), \quad j \in \mathbb{Z}.$$

Removing two consecutive corner points cuts a lifted period into two open arcs. Define

$$T(\gamma, \theta, \alpha, \beta) = (\gamma + 21\pi, \theta + 10\pi, \alpha, \beta + 7\pi).$$

Modulo the parameter period lattice, T is an involution exchanging the two arcs. With $\rho = \tau\sigma$, the boundary maps satisfy

$$r_1 T = \rho r_1, \quad r_2 T = \rho r_2, \quad r_3 T = \sigma r_3. \quad (7)$$

The inherited primitive F may be normalized so that $F \circ T = F$.

Let I_0 be the open arc $0 < \gamma < 21\pi$, let $I_1 = T(I_0)$, and define

$$\text{can}(z) = z \quad (z \in I_0), \quad \text{can}(z) = T^{-1}z \quad (z \in I_1).$$

For every root $x = (z_-, z_+)$, including a root whose preimages lie on different open arcs,

$$\mathcal{A}_t(x) = \int_{\text{can}(z_+)}^{\text{can}(z_-)} r_3^* \lambda, \quad (8)$$

where the integral follows the oriented subarc of I_0 .

Proof. At a parameter corner (g, h, k) , the three boundary characters are (g, k) , $(g, k + h)$, and (g, h) . For odd q , a corner (g, r) lies on $A_{1/q}$ precisely when $r = g$. The two input equations therefore give $k = g$ and $h = 0$, which yields exactly the displayed corner lifts. Their consecutive spacing shows that each full period contains two corner cuts.

The square of T is the parameter period $(42\pi, 20\pi, 0, 14\pi)$. Direct substitution in the defining traceless equation shows that T preserves the parameter space, and substitution in the boundary words gives (7). The map ρ preserves both odd-slope arcs. Average the primitive on each rational arc under ρ and the primitive on C_t under T . The normalized Liouville form is invariant under τ and σ , so the averages have the same exterior derivatives and the composition primitive satisfies $F \circ T = F$.

On either open component, $dF = r_3^* \lambda$. Replace both endpoints by their representatives in I_0 and integrate from the positive endpoint to the negative endpoint. The interior of I_0 contains no all-corner point, so no residue detour is involved and (8) follows. $\square$

3. THE SIX FOLDS AND THE SEPARATED GRAPH SHEETS

The main part of the partial composition

$$F_t^{\text{main}} = A_{1/3} \times_{P_1} C_t \looparrowright P_2^- \times P_3$$

is used here only to describe the finite geometry of the direct output. No associativity or curved-composition theorem is invoked.

Proposition 3.1 (Six-circle and fold table). *The singular main partial composition has six stabilizer circles. Their P_2 seam values, the corresponding $A_{1/3}$ angle θ_1 , and the image interval $[\ell_i, h_i]$ in the third pillowcase are*

i	(γ_2, θ_2)	θ_1	$[\ell_i, h_i]$
1	$(0, 2\pi/7)$	$2\pi/3$	$[8\pi/21, 20\pi/21]$
2	$(0, 4\pi/7)$	$2\pi/3$	$[2\pi/21, 16\pi/21]$
3	$(0, 6\pi/7)$	$2\pi/3$	$[4\pi/21, 10\pi/21]$
4	$(\pi, \pi/7)$	$\pi/3$	$[4\pi/21, 10\pi/21]$
5	$(\pi, 3\pi/7)$	$\pi/3$	$[2\pi/21, 16\pi/21]$
6	$(\pi, 5\pi/7)$	$\pi/3$	$[8\pi/21, 20\pi/21]$

If ψ parametrizes a stabilizer circle, then

$$\cos \theta_3(\psi) = \cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2 \cos \psi. \quad (9)$$

Both endpoints of every interval are nondegenerate folds. The pillowcase symmetry pairs rows (1, 6), (2, 5), and (3, 4), including their fold coefficients.

Proof. A lift of $A_{1/q}$ satisfies $q\theta - \gamma \in 2\pi\mathbb{Z}$. Its noncorner seam values are

$$A_{1/3} : (0, 2\pi/3), (\pi, \pi/3)$$

and

$$A_{1/7} : (0, 2\pi k/7), (\pi, (2k-1)\pi/7), \quad k = 1, 2, 3.$$

Pairing the unique $A_{1/3}$ value on each seam with the three $A_{1/7}$ values gives the six circles. Choose the common seam axis as polar axis. The Euclidean dot product of the two remaining axes gives (9), and hence

$$[\ell_i, h_i] = [|\theta_1 - \theta_2|, \pi - |\pi - (\theta_1 + \theta_2)|].$$

At the two endpoints,

$$\begin{aligned} \theta_3(\psi) &= \ell_i + \frac{\sin \theta_1 \sin \theta_2}{2 \sin \ell_i} \psi^2 + O(\psi^4), \\ \theta_3(\pi + u) &= h_i - \frac{\sin \theta_1 \sin \theta_2}{2 \sin h_i} u^2 + O(u^4). \end{aligned}$$

All coefficients are positive and finite. The symmetry assertion follows by replacing the two input angles by their complements. $\square$

Proposition 3.2 (Hyperbolic smoothing at the twelve endpoints). *Use the group-presentation convention*

$$p = \exp(t \operatorname{Im}(ac^{-1}x)), \quad \Phi_t = \operatorname{Re}(xp^{-1}a^{-1}pb),$$

with

$$a = \mathbf{i}, \quad b = e^{\gamma \mathbf{k}} \mathbf{i}, \quad c = e^{\theta \mathbf{k}} \mathbf{i}, \quad x = \sin \alpha \cos \beta \mathbf{i} + \sin \alpha \sin \beta \mathbf{j} + \cos \alpha \mathbf{k}.$$

Near the lower and upper endpoints of the i th circle there are smooth coordinates (u, v) in which

$$\{\Phi_t = 0\} = \{uv = \tau_i^\pm(t)\}, \quad \tau_i^\pm(0) = 0, \quad (\tau_i^\pm)'(0) = \mp 2 \sin \theta_1 \sin \theta_2.$$

Thus all twelve endpoints have nondegenerate hyperbolic smoothing germs, with opposite sectors at the two ends of each circle and the same three symmetry pairs as above.

Proof. At $t = 0$ the defining equation is

$$\Phi_0(\gamma, \theta, \alpha, \beta) = \sin \gamma \cos \alpha.$$

Write $X = \gamma - \gamma_i$ and $Y = \alpha - \pi/2$, where $\gamma_i \in \{0, \pi\}$ and $\epsilon_i = \cos \gamma_i$. Quaternion multiplication gives

$$\Phi_t = \epsilon_i(-XY - 2t \sin \beta \sin(\theta - \beta)) + O(t(|X| + |Y|) + t^2 + (|X| + |Y|)^3).$$

The Hessian in (X, Y) is nonsingular with index one. The parametric Morse lemma therefore gives $uv = \tau(t)$ and

$$\tau'(0) = -2 \sin \beta \sin(\theta - \beta).$$

At the lower and upper lifts the products are respectively $-\sin \theta_1 \sin \theta_2$ and $\sin \theta_1 \sin \theta_2$, proving the formulas. $\square$

Proposition 3.3 (Off-seam graph sheets). *Away from neighborhoods of the six seam values, every branch of F_0^{main} is locally the graph of the symplectic shear*

$$\phi_c(\gamma, \eta) = (\gamma, \eta + \gamma/3 + c).$$

On a torus-cover chart, the 21 local lifts of the direct composition are parallel and pairwise disjoint. Both statements persist on compact off-seam subsets for all sufficiently small $t > 0$.

Proof. On the unperturbed main stratum, the three boundary coordinates are

$$p_1 = (\gamma, \beta), \quad p_2 = (\gamma, \theta - \beta), \quad p_3 = (\gamma, \theta).$$

The first rational arc gives $\beta = \gamma/3 + c$. With $\eta = \theta - \beta$, this is the displayed shear, and $d\gamma \wedge d(\eta + \gamma/3 + c) = d\gamma \wedge d\eta$. By Smith's cross-resolution theorem, the main subspace is obtained from these graph branches by replacing neighborhoods of the six internal circle fibers with their local resolutions [Smi24, Thm. 4.22]. Thus every failure of the graph description within the main subspace is contained in those six neighborhoods. The auxiliary circles allowed by Smith's theorem are separate components of the full fiber product and are not part of F_t^{main} .

Choose lifts

$$\beta = \frac{\gamma + 2\pi k}{3}, \quad \eta = \frac{\gamma + 2\pi \ell}{7}, \quad (k, \ell) \in \mathbb{Z}/3 \times \mathbb{Z}/7.$$

Then

$$\theta = \frac{10}{21}\gamma + \frac{2\pi(7k + 3\ell)}{21}.$$

The homomorphism $(k, \ell) \mapsto 7k + 3\ell$ from $\mathbb{Z}/3 \times \mathbb{Z}/7$ to $\mathbb{Z}/21$ is bijective, so the 21 lifts have distinct intercepts. A compact off-seam set has positive minimum separation, which persists under the C^1 small- t deformation. $\square$

4. THE EXACT ROOT-ACTION TABLE

Use scaled coordinates on the canonical arc I_0 :

$$s = \frac{\gamma}{\pi}, \quad n = \frac{21\theta}{\pi}.$$

A vertical torus period changes n by 42, and the arc runs from $(0, 0)$ to $(21, 210)$.

Lemma 4.1 (Singular first-arc data). *Between consecutive integral values of s , the singular first arc has slope $d\theta/d\gamma = 10/21$. Its twelve signed vertical jumps are*

s	i	Δn	n -interval	s	i	Δn	n -interval
1	4	-6	[4, 10]	11	5	14	[110, 124]
4	1	-12	[22, 34]	12	1	12	[134, 146]
5	4	6	[32, 38]	14	2	-14	[152, 166]
7	5	-14	[44, 58]	16	3	6	[172, 178]
9	6	12	[64, 76]	17	6	-12	[176, 188]
10	2	14	[86, 100]	20	3	-6	[200, 206]

At the other interior integral seams, the diagonal sheet has the eight points

$$(2, 14), (3, 24), (6, 48), (8, 54), (13, 156), (15, 162), (18, 186), (19, 196).$$

Proof. Proposition 3.3 gives $\theta = \gamma/3 + \gamma/7 + c$, so increasing s by one increases n by ten. At an integral seam, substitute the six intervals of Proposition 3.1 and choose the lift whose incoming endpoint agrees with the preceding diagonal segment. Continuing through the outgoing endpoint gives the twelve jumps. The recurrence

$$n_j^- = n_{j-1}^+ + 10, \quad n_j^+ = n_j^- + \Delta n_j$$

gives all displayed intervals and point values. The jumps sum to zero, so the endpoint is $(21, 210)$. $\square$

Lemma 4.2 (Algebraic normal roots). *Put $x = \cos \theta$ and $u = 2 \cos(\pi/7)$, so*

$$u^3 - u^2 - 2u + 1 = 0.$$

On row i of Proposition 3.1, the small- t normal displacement is

$$\gamma = j\pi + tG_i(\theta) + O(t^2), \quad G_i(\theta) = \frac{2(a_i - \cos \theta)}{\cos \alpha_i(\theta)}, \quad (10)$$

where

$$\cos^2 \alpha_i(\theta) = \frac{b_i^2 - (\cos \theta - a_i)^2}{\sin^2 \theta}, \quad \text{sgn}(\cos \alpha_i) = \begin{cases} +1, & i = 1, 2, 3, \\ -1, & i = 4, 5, 6, \end{cases} \quad (11)$$

and

$$a_1 = a_6 = \frac{2 - u^2}{4}, \quad a_2 = a_5 = \frac{u^2 - u - 1}{4}, \quad a_3 = a_4 = \frac{u}{4},$$

$$b_i^2 = \frac{3}{4}(1 - 4a_i^2).$$

The physical simple solutions of $G_i = G_j$ on $0 < \theta < \pi$ are exactly

root	row pairs $\{i, j\}$	x -box	$21\theta/\pi$ -box
A	$\{1, 4\}, \{3, 6\}$	$[0.178447, 0.178449]$	$[9.300, 9.301]$
B	$\{1, 5\}, \{2, 6\}$	$[-0.123491, -0.123489]$	$[11.327, 11.328]$
C	$\{1, 6\}$	$[-0.311746, -0.311743]$	$[12.619, 12.620]$
D	$\{2, 3\}, \{4, 5\}$	$[0.722519, 0.722522]$	$[5.102, 5.103]$
E	$\{2, 4\}, \{3, 5\}$	$[0.346009, 0.346012]$	$[8.138, 8.139]$
F	$\{2, 5\}$	$[0.111259, 0.111262]$	$[9.754, 9.755]$
G	$\{3, 4\}$	$[0.450483, 0.450486]$	$[7.376, 7.377]$.

There is no physical root for the row pairs $\{1, 2\}, \{1, 3\}, \{4, 6\}, \{5, 6\}$.

Proof. At $t = 0$, substitution of the seam rational-arc equations into the quaternionic boundary words gives (11). Differentiating the defining equation in the normal γ direction and in t gives

$$\partial_\gamma \Phi_0 = (-1)^j \cos \alpha_i, \quad \partial_t \Phi_0 = -2(-1)^j (a_i - \cos \theta),$$

which proves (10). The formulas for a_i and b_i follow from the minimal polynomial of u .

For pairs of distinct symmetry types, squaring $G_i = G_j$ and clearing the positive fold denominators gives the three quadratics $P_{rs}(x) = A_{rs}x^2 + B_{rs}x + C_{rs}$ with

$$(A_{12}, B_{12}, C_{12}) = \left(u^2 - 2u - 1, -\frac{5}{2}u^2 + \frac{3}{2}u + \frac{5}{2}, \frac{1}{4}u^2 - \frac{1}{2}u - \frac{1}{4} \right),$$

$$(A_{13}, B_{13}, C_{13}) = \left(u^2 - \frac{1}{2}u - \frac{3}{2}, -\frac{3}{4}u^2 - \frac{1}{2}u + 2, \frac{1}{4}u^2 - \frac{1}{8}u - \frac{3}{8} \right),$$

$$(A_{23}, B_{23}, C_{23}) = \left(u^2 + u - 2, u^2 - \frac{5}{2}u - 2, \frac{1}{4}u^2 + \frac{1}{4}u - \frac{1}{2} \right).$$

The rational interval

$$1.801937735 < u < 1.801937737$$

contains $2 \cos(\pi/7)$ and one simple zero of $u^3 - u^2 - 2u + 1$. Rational interval evaluation gives one physical root of P_{12} , one of P_{13} , and both roots of P_{23} in the displayed boxes. Each polynomial has opposite strict signs at the endpoints of its box and a derivative of one strict sign there. Vieta's formula places the unused root of P_{12} below -1 and that of P_{13} above 1 . The signs in (11) remove the extraneous squared solutions. For equal symmetry types the two functions agree only at $x = a_i$, producing C, F, G, and their derivatives are nonzero.

The angle boxes admit rational interval proofs. Machin's identity

$$\frac{\pi}{4} = 4 \arctan \frac{1}{5} - \arctan \frac{1}{239}$$

and alternating sums give a rational π -interval of width less than 1.84×10^{-35} . Rational Taylor bounds for cosine then place every angle strictly inside its displayed box and determine the derivative signs. $\square$

Lemma 4.3 (Area-residue identity on the torus cover). *Use scaled universal-cover coordinates*

$$x = \gamma/\pi, \quad y = \theta/\pi.$$

Write $\tilde{\lambda}_0$ for the lift of the normalized primitive from Lemma 2.1. The deleted orbifold points lift to $\mathbb{Z}^2$, the lifted symplectic form is $\pi^2 dx \wedge dy$, and $\tilde{\lambda}_0$ has limiting period $-\pi^2$ around every positively oriented small meridian of a point in $\mathbb{Z}^2$. Hence every piecewise smooth closed curve $\Gamma \subset \mathbb{R}^2 \setminus \mathbb{Z}^2$ satisfies

$$\frac{1}{\pi^2} \int_{\Gamma} \tilde{\lambda}_0 = \text{Area}(\Gamma) - \sum_{z \in \mathbb{Z}^2} \text{wind}_z(\Gamma), \quad (12)$$

where $\text{Area}(\Gamma) = \int_{\mathbb{R}^2} \text{wind}_q(\Gamma) dq$ is signed area.

Suppose a path segment has deck-related endpoints. Concatenate it with its image under the pillowcase involution and join the remaining endpoints by an involution-paired choice of closing arcs. If a and w denote the signed area and total lattice winding of the resulting closed curve, then the original segment has integral $\pi^2(a - w)/2$. If the segment must first be doubled to obtain deck-related endpoints, its integral is $\pi^2(a - w)/4$.

Proof. The torus cover has Goldman area $4\pi^2$ and is generically two-to-one over the pillowcase, so the pillowcase area is $2\pi^2$. By Lemma 2.1, the four positively oriented meridian periods are equal. Their sum is the negative of the total area by Stokes' theorem, so each is $-\pi^2/2$. A small meridian about a lattice point in the torus cover maps with degree two to a pillowcase meridian; its lifted period is therefore $-\pi^2$.

Apply Stokes' theorem to the winding-number two-chain bounded by Γ , with small disks about the lattice points removed. The area term is $\pi^2 \text{Area}(\Gamma)$, and a puncture with winding m contributes $m\pi^2$ from the clockwise boundary of its deleted disk. Moving these boundary terms to the other side gives (12).

The normalized primitive is invariant under the pillowcase involution. The two closing arcs can be chosen as involution images with opposite orientations, so their integrals cancel, while the original segment and its involuted image have equal integrals. The closed curve therefore has twice the integral of the segment. If an initial deck-doubling was required, the doubled segment has twice the original integral once more. This proves both factors. $\square$

Proposition 4.4 (Complete limiting action table). *The interval data of Lemma 4.1 have 58 pairwise seam overlaps. Within the compact main output, exactly 50 persist as transverse degree-one roots for all sufficiently small $t > 0$: 28 are diagonal-circle roots and 22 are the simple circle-circle roots of Lemma 4.2. The eight excluded seam pairs are*

$$\{1, 17\}, \{4, 14\}, \{4, 20\}, \{5, 9\}, \{7, 17\}, \{9, 11\}, \{10, 12\}, \{12, 16\}.$$

The following table gives one representative of each half-period pair. Here $d \in 42\mathbb{Z}$ is the offset in n , r is a normal solution or pt for a diagonal-circle root, D is the short deck class defined above, and o is the degree-one ordering. The columns a, w, κ are the area, winding, and doubling factor of Lemma 4.3; thus

$$\frac{1}{\pi^2} \int_j^k \lambda = \frac{\kappa}{2}(a - w). \quad (13)$$

$(j, k) \sim (j', k')$	d	r	D	o	a	w	κ	$\mathcal{A}/\pi^2$
$(1, 6) \sim (15, 20)$	42	pt	(8, 4)	$1 \rightarrow 6$	80/21	2	1/2	-19/42
$(1, 7) \sim (14, 20)$	42	D	(3, 1)	$7 \rightarrow 1$	-8/7	-2	1	3/7
$(1, 10) \sim (11, 20)$	84	E	(6, 3)	$1 \rightarrow 10$	124/7	14	1/2	-13/14
$(1, 12) \sim (9, 20)$	126	A	(5, 2)	$1 \rightarrow 12$	288/7	38	1/2	-11/14
$(1, 16) \sim (5, 20)$	168	G	(3, 1)	$1 \rightarrow 16$	36	34	1/2	-1/2
$(2, 7) \sim (14, 19)$	42	pt	(8, 4)	$7 \rightarrow 2$	24/7	2	1/2	5/14
$(2, 10) \sim (11, 19)$	84	pt	(4, 2)	$2 \rightarrow 10$	96/7	12	1	-6/7
$(2, 12) \sim (9, 19)$	126	pt	(5, 3)	$2 \rightarrow 12$	192/7	26	1	-5/7
$(2, 17) \sim (4, 19)$	168	pt	(3, 1)	$17 \rightarrow 2$	244/7	34	1/2	3/14
$(3, 4) \sim (17, 18)$	0	pt	(10, 5)	$4 \rightarrow 3$	-116/21	-6	1/2	5/42
$(3, 9) \sim (12, 18)$	42	pt	(3, 1)	$3 \rightarrow 9$	-16/21	-2	1	-13/21
$(4, 5) \sim (16, 17)$	0	A	(10, 5)	$4 \rightarrow 5$	-36/7	-6	1/2	-3/14
$(4, 9) \sim (12, 17)$	42	C	(8, 4)	$4 \rightarrow 9$	8	6	1/2	-1/2
$(4, 11) \sim (10, 17)$	84	B	(7, 3)	$4 \rightarrow 11$	284/7	38	1/2	-9/14
$(4, 13) \sim (8, 17)$	126	pt	(6, 2)	$4 \rightarrow 13$	1496/21	70	1/2	-13/42
$(5, 11) \sim (10, 16)$	84	D	(3, 2)	$5 \rightarrow 11$	188/7	26	1	-3/7
$(5, 14) \sim (7, 16)$	126	E	(6, 2)	$14 \rightarrow 5$	488/7	70	1/2	-1/14
$(5, 15) \sim (6, 16)$	126	pt	(5, 3)	$15 \rightarrow 5$	544/21	26	1	-1/21
$(6, 7) \sim (14, 15)$	0	pt	(10, 5)	$7 \rightarrow 6$	-212/21	-10	1/2	-1/42
$(6, 10) \sim (11, 15)$	42	pt	(2, 1)	$6 \rightarrow 10$	188/21	8	1	-10/21
$(7, 8) \sim (13, 14)$	0	pt	(10, 5)	$7 \rightarrow 8$	-28/3	-10	1/2	-1/6
$(7, 10) \sim (11, 14)$	42	F	(9, 4)	$7 \rightarrow 10$	32	30	1/2	-1/2
$(7, 12) \sim (9, 14)$	84	B	(8, 3)	$7 \rightarrow 12$	500/7	70	1/2	-5/14
$(8, 10) \sim (11, 13)$	42	pt	(1, 1)	$8 \rightarrow 10$	68/3	22	1	-1/3
$(8, 12) \sim (9, 13)$	84	pt	(2, 2)	$8 \rightarrow 12$	932/21	44	1	-4/21.

Thus 42 roots have negative limiting action and 8 have positive limiting action. Every limiting action is nonzero, and the minimum absolute value is $\pi^2/42$.

Proof. For seams $j < k$, canonicalized output points agree precisely when their n -intervals overlap after a shift $d \in 42\mathbb{Z}$. Intersecting the twenty intervals in Lemma 4.1 gives 58 nonempty triples (j, k, d) . Twenty-eight contain one diagonal point strictly inside a vertical interval. The other thirty are circle-circle overlaps. Substitution of the isolated normal solutions from Lemma 4.2 leaves one simple root in 22 and none in the eight listed pairs. The strict angle boxes lie inside their overlap intervals.

At a diagonal-circle root, the limiting directions in scaled coordinates are $(1, 10/21)$ and $(0, \Delta n_j)$. At a circle-circle root in rows i, i' , the determinant has leading sign

$$\operatorname{sgn}(\Delta n_j \Delta n_k) \operatorname{sgn}(G'_i(\theta) - G'_{i'}(\theta)).$$

The rational derivative bounds therefore fix the degree-one ordering and prove transversality. Choose mutually disjoint neighborhoods of the 50 simple roots and of the twelve folds. Every root neighborhood lies a positive distance from a fold because the diagonal points and the angle boxes are strict interior points of their overlap intervals. On the complement of these neighborhoods in the compact main domain, Proposition 3.3 gives a positive minimum separation between every pair of output branches. This separation persists under the C^1 small- t deformation. In a fold neighborhood, Proposition 3.2 identifies the entire local main resolution with one embedded hyperbola. The implicit-function theorem continues each of the 50 simple roots uniquely. These sets cover the compact main domain, so there are no additional roots there.

For an overlap (j, k, d) choose a common height h and put

$$p = (j, h/21), \quad q = (k, (h + d)/21)$$

in $(\gamma/\pi, \theta/\pi)$ coordinates. Follow the singular path from p to q . If $k - j$ is even, its endpoints differ by an ordinary torus deck transformation and $\kappa = 1$. If $k - j$ is odd, concatenate the path with its endpoint translate and divide the resulting integral by two; then $\kappa = 1/2$.

All paths lie in P^δ , where $\lambda = \lambda_0$ by Lemma 2.1. Apply Lemma 4.3, using involution-paired closing arcs. It gives

$$\frac{1}{\pi^2} \int_p^q \lambda = \frac{\kappa}{2} \left(\text{Area} - \sum_{z \in \mathbb{Z}^2} \text{wind}_z \right),$$

which is (13). Shoelace evaluation on the vertices of Lemma 4.1 gives the printed a and w . The common height can move inside its overlap interval without crossing a lattice point, so the integral is constant and may be evaluated at a rational midpoint even when the actual normal root is algebraic.

For example, $(1, 7)$ has $(a, w, \kappa) = (-8/7, -2, 1)$, so the $7 \rightarrow 1$ action is $3\pi^2/7$. The pair $(3, 4)$ has $(a, w, \kappa) = (-116/21, -6, 1/2)$, so the $4 \rightarrow 3$ action is $5\pi^2/42$. The half-period map sends

$$(j, k, d) \mapsto (21 - k, 21 - j, d)$$

and preserves the normalized action, ordering, and short deck class. Each row therefore has multiplicity two. Reading the last column proves the sign count and the minimum gap. $\square$

Corollary 4.5 (Small-perturbation persistence). *There is $t_0 > 0$ such that, for every $0 < t < t_0$, the compact main output L_t^δ has exactly the 50 roots in Proposition 4.4, and all have the signs shown there.*

Proof. Transversality continues all 50 roots uniquely. Compact separation of the graph sheets and the fold neighborhoods excludes new roots. Proposition 2.3 gives primitives that vary continuously with t , and the canonical paths remain in the compact corner-free arc I_0 . Thus every action converges to its table value. Since the finite minimum absolute limiting value is $\pi^2/42$, every sign is constant after decreasing t_0 . $\square$

Proof of Theorem 1.1. Exactness is Proposition 2.2, and the relative primitive is fixed by Proposition 2.3. Proposition 4.4 proves the complete root and limiting-action count. Corollary 4.5 proves persistence for small positive t . $\square$

5. ENERGY INTERPRETATION AND SCOPE

The sign table has an elementary consequence for any exact holomorphic configuration that may be defined using L_t^δ . The statement does not assume that a compactified figure-eight operation exists.

Lemma 5.1 (Root-action identity). *Let a finite tree of exact pseudoholomorphic or Morse components have no external inputs and one external root x on L_t^δ . Suppose its seam and boundary lifts are fixed and every component has the usual Stokes energy identity. Then*

$$E_{\text{tot}} = \mathcal{A}_t(x).$$

Consequently, a nonconstant such configuration can be rooted only at a positive-action root.

Proof. Apply Stokes' theorem to each component. At every internal flag the same primitive difference appears with opposite signs on the two adjacent components. The action drops along Morse edges cancel in the same way. With no external inputs, only the external output remains. Compatible almost-complex and Morse data make each energy summand nonnegative, and a nonconstant component makes the total positive. $\square$

Corollary 5.2. *For $0 < t < t_0$, every nonconstant exact zero-input configuration satisfying the hypotheses of Lemma 5.1 is rooted at one of the eight positive roots in Theorem 1.1.*

Remark 5.3 (No counting conclusion). Corollary 5.2 is an action obstruction, not an existence or parity theorem. It does not assert that any of the eight positive roots supports a holomorphic configuration, that only one vertex occurs, or that a sum of outputs defines a bounding cochain. Those conclusions would require compactification, transversality, gluing, local mod-two counts, and continuation data that are not constructed here.

Remark 5.4 (Reproducibility). In the arXiv source archive, the command

```
python3 anc/q7_exact_actions.py
```

reconstructs the singular first-arc data, verifies the rational root enclosures and derivative signs, enumerates all 58 overlaps, and prints the 25-row area–residue table using exact rational arithmetic. The standard-library program reproduces the finite calculations; the proof does not depend on its execution.

Theorem 1.1 therefore supplies the complete finite action filtration of the compact main output. It does not extend that filtration to auxiliary components or the deleted end collars. Holomorphic and categorical questions remain outside the claimed result.

REFERENCES

- [BW18] N. Bottman and K. Wehrheim, *Gromov compactness for squiggly strip shrinking in pseudoholomorphic quilts*, *Selecta Math. (N.S.)* **24** (2018), 3381–3443. arXiv:1503.03486.
- [CHKK22] G. Cazassus, C. Herald, P. Kirk, and A. Kotelskiy, *The correspondence induced on the pillowcase by the earring tangle*, *J. Topol.* **15** (2022), 2472–2543. arXiv:2010.04320.
- [CHK22] G. Cazassus, C. Herald, and P. Kirk, *Tangles, relative character varieties, and holonomy perturbed traceless flat moduli spaces*, *Open Book Ser.* **5** (2022), 95–121. arXiv:2102.00901.
- [HK18] C. Herald and P. Kirk, *Holonomy perturbations in a cylinder, and regularity for traceless $SU(2)$ character varieties of tangles*, *Geom. Topol.* **22** (2018), 129–216. arXiv:1511.00308.
- [Smi24] K. Smith, *Perturbed traceless $SU(2)$ character varieties of tangle sums*, arXiv:2412.06066.
- [Wue26] B. J. Wuebben, *The instanton homology of the $(-2, 3, q)$ pretzel knots and computed bounding cochains in the pillowcase*, arXiv:2607.26096.

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